GOOGLE CLOUD PLATFORM: AN INSIDER'S GUIDE

By Conner Forrest

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TechRepublic.

INTRODUCTION

From the humble beginnings with Google App Engine back in 2008, Google has grown its Google Cloud Platform (GCP) into one of the premier cloud computing platforms on the market today. While it is still following its top competitors, Amazon Web Services (AWS) and Microsoft Azure, Google is holding its own in the cloud wars and continues to make investments in GCP that make the product more attractive to big customers.

To help CXOs, IT leaders, operations administrators, and developers better understand Google's role as a cloud provider, we've put together the most important details and resources in this ebook.

EXECUTIVE SUMMARY (TL;DR)

- **What is Google Cloud Platform?** Google Cloud Platform, as the name implies, is a cloud computing platform that provides infrastructure tools and services for users to build on top of.
- **Why does Google Cloud Platform matter?** Google Cloud Platform is regarded as the third biggest cloud provider in terms of revenue, behind AWS in first place and Microsoft Azure in second.
- **Who does Google Cloud Platform affect?** Any organization in need of cloud computing should consider Google Cloud Platform—especially SMBs, which the platform was initially geared toward.
- **When was Google Cloud Platform announced?** Google announced its first cloud tool, Google App Engine, back in 2008. It continued to add more tools and services until it eventually became known as the Google Cloud Platform.
- **How can I use Google Cloud Platform?** Google has provided documentation for getting started and a frequently asked questions page for developers and IT leaders to investigate the platform.

WHAT IS GOOGLE CLOUD PLATFORM?

In 2008, to capture the growing interest in web applications, Google launched Google App Engine, a platform-as-a-service (PaaS) cloud tool that allowed developers to build and host their apps on Google's infrastructure. App Engine struggled early on because it didn't support certain key developer languages.

Google then released a host of complementary tools, such as its data storage layer and its infrastructure-as a-service (IaaS) component, known as the Google Compute Engine, which supports the use of virtual machines. After growing as an IaaS provider, Google added more products, including a load balancer, DNS, monitoring tools, and data analysis services, making GCP better able to compete in the cloud market and increasing its market share.

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Current GCP products span the following 13 categories:

- Compute—App Engine, Compute Engine, Kubernetes Engine, Cloud Functions (beta)
- Storage & Databases—Cloud Storage, Cloud Bigtable, Cloud SQL, Cloud Datastore, and more
- Networking—Virtual Private Cloud (VPC), Cloud Load Balancing, Network Service Tiers, Cloud Armor, and more
- Big Data—BigQuery, Cloud Dataflow, Cloud Dataproc, Cloud Pub/Sub, and more
- Cloud AI—Cloud Machine Learning Engine, Cloud TPU, Cloud AutoML (beta), various machine learning APIs
- Identity & Security—Cloud Identity, Cloud IAM, Security Key Enforcement, Cloud Security Scanner, Cloud Resource Manager, and more
- Management Tools—Stackdriver Overview, Monitoring, Trace, Logging, Debugger, Cloud Console, and more
- Developer Tools—Cloud SDK, Container Registry, Container Builder, Cloud Test lab, and more
- API Platform and Ecosystems—Google Maps Platform, API Analytics, API Monetization, Cloud Endpoints, and more
- Data Transfer—Google Transfer Appliance, Cloud Storage Transfer Service, Google BigQuery Transfer Service
- Productivity Tools—G Suite, Hire, Chrome, Android
- Professional Services—Consulting, Technical Account Management, Training, Certification, and more
- Internet of Things—Cloud IoT Core

GCP is primarily a public cloud provider. Google does have a network of private cloud providers that can help users build out a hybrid cloud deployment, but its proprietary space is the public cloud. The platform also has a host of other partners that provide additional services.

While AWS and Microsoft consistently push each other to lower prices, Google follows its own pricing model and routinely boasts that it offers the lowest cost of the three providers. However, Google really differentiates itself in its services.

Read more

- The Google Cloud Platform: 10 things you need to know (TechRepublic)
- Google's master cloud plan: Buy more infrastructure, charge less for it (TechRepublic)
- Why machine learning is so critical to the future of Google Cloud (TechRepublic)
- Cloud migration decision tool (Tech Pro Research)

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WHY DOES GOOGLE CLOUD PLATFORM MATTER?

Whether Google Cloud Platform will matter to your organization depends on the type of tools and functionality it values from a cloud provider.

In terms of basic services, Google offers about the same core functionalities of AWS and Azure, but on a smaller scale. Where it really shines, though, is in its big data tools, artificial intelligence (AI) and machine learning initiatives, and container support.

Google's BigQuery and Dataflow bring strong analytics and processing capabilities for companies that work heavily with data, while Google's Kubernetes container technology allows for clear container cluster management and eases container deployment. Google's Cloud Machine Learning Engine and various machine learning APIs make it easier for businesses to leverage AI in the cloud.

Google is a company that thrives on the collection and subsequent leveraging of data. Whether that is user data, machine data, or geographic data is irrelevant—if an enterprise wants to experiment with data, GCP may be a good option as a cloud provider.

In addition to its work in AI and machine learning, GCP stays on top of developing enterprise trends like serverless computing to remain competitive against tools such as AWS Lambda. GCP boasts serverless solutions across app development, analytics, and more. It also has an integration with Elastic Cloud to support open source search and analytics.

Google Cloud Platform also matters because of the massive investment Google is making in its infrastructure. As noted by TechRepublic columnist Matt Asay, in 2014 Google spent more than AWS and Microsoft combined on its cloud infrastructure.

Those investments are seen most clearly in what Google perceives as its three keys to success in the future: Machine learning, data, and containers. At the 2017 Google Cloud Cloud Next conference in San Francisco, Google leaders explained how they were working in these areas, as well as compute and security, to make Google Cloud Platform a better option for enterprise customers and an attractive cloud hosting solution.

Read more

- How Google Cloud Platform supports IoT development (TechRepublic)
- How developers can take advantage of machine learning on Google Cloud Platform (TechRepublic)
- Google Cloud Platform's 3 keys to the roadmap: Data center, security, containers (TechRepublic)
- Google Cloud Platform adds more managed database services (TechRepublic)

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WHO DOES GOOGLE CLOUD PLATFORM AFFECT?

As with many of Google's innovations, the set of tools that GCP comprises were originally internal tools built for Google's use. This would eventually prove problematic.

At the outset, Google targeted startups and SMBs for its cloud division. The company even went as far as to create a startup fund, offering $100,000 of Cloud Platform credits to eligible startups in 2014. However, Google would eventually need to expand its focus to prove successful as a cloud provider.

The problem was, Google marketed its products for users to be able to build their apps just like Google did. But many companies, especially bigger enterprises, didn't want to. Google executive chairman Eric Schmidt admitted in 2016 that this was the wrong approach and that Google had decided to change gears in how it provides its services through GCP.

"We decided to meet you where you are, as opposed to where we think you should be," Schmidt said at the time.

By signing former VMware honcho Diane Greene, adding security and reliability, and providing better stepping stones into GCP, Google is making its platform more accessible to all businesses looking to ditch some (or all) of their data centers. So where GCP would have once affected only small businesses and startups, it's now a viable option for enterprises and big business workloads as well.

It seems that large corporations are paying attention. In 2016, Google added big names to its Cloud Platform roster, including Disney, Coca-Cola, Spotify, Apple, Colgate-Palmolive, and Home Depot, proving that it can cater to the needs of major players. However, many of these companies also use other providers, such as AWS or Azure, which means that GCP could act as a complementary provider for existing AWS or Azure customers who need additional capabilities or flexibility.

Since it is a platform on which applications are built and hosted, the choice of a provider like GCP also affects developers. For them, GCP supports Go, Node.JS, Python, Ruby, PHP, .NET, and Java. Developers should make sure they are involved in any conversations about selecting a cloud provider to ensure that it is a platform they and their team are comfortable working in.

Read more

- Here's what developers really think about AWS, Microsoft Azure, and Google Cloud (TechRepublic)
- Google Cloud Platform sets up new global startup fund (ZDNet)
- Google Cloud touts major enterprise customers as it runs to catch up to AWS (ZDNet)
- Google admits original enterprise cloud strategy was wrong, why it's gone in a different direction (TechRepublic)

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- Google Cloud Platform signs up enterprise giants, how does it compare to AWS? (TechRepublic)
- Google Cloud Platform courts enterprises and they'll bite on tried and true pitches (ZDNet)

WHEN WAS GOOGLE CLOUD PLATFORM ANNOUNCED?

As mentioned, Google's first foray into cloud services was the Google App Engine back in 2008. Two years later, Google announced that it was adding a storage layer, and in 2012, the company began its partner program for the platform. Then came BigQuery, the Compute Engine, Cloud SQL, and the rest of the tools that make up today's Google Cloud Platform.

Most of the GCP products mentioned above are in general availability now. However, like all providers, Google is constantly adding new tools and features in preview or beta, which will likely make it to the general public.

Some of the latest tools brought to the Cloud Platform are Cloud Job Discovery, Cloud Text-to-Speech, Access Transparency, Cloud Security Command Center, Cloud Functions, and Cloud Armor. In addition to its existing set of APIs and Machine Learning Engine, Google launched its Cloud TPU beta to speed up efforts of machine learning on workloads and offer on-demand supercomputing.

Read more

- Developers, start your engines (Google)
- Google Cloud Platform adds regions, aims to close global cloud computing gap (ZDNet)
- Google intros Compute Engine infrastructure service (ZDNet)
- Google quietly launches GitHub cloud rival (ZDNet)
- Why machine learning and data analysis are critical to Google's success in the cloud (TechRepublic)

HOW CAN I USE GOOGLE CLOUD PLATFORM?

Since GCP is a publicly available product, it's not difficult to acquire its services. The bigger issue is twofold: deciding whether the platform is the best option for your business and planning your migration.

To effectively compare GCP against the other options out there, you'll need to do your research. If you are comparing it against AWS and Microsoft Azure, try starting with our guides for AWS and Microsoft Azure. A list of other good cloud vendors can be found in this list of 15 of the top hybrid cloud vendors.

As your organization begins to plan its deployment, start by making a list of questions you have about the service and check them against the FAQ section on the GCP website. To understand the specifics of a GCP deployment, make sure you familiarize yourself with the proper documentation.

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Google does offer a free tier for GCP, as well as a free 12-month trial with credit for organizations that may need to dip their toes in the water. A tool for live migrations is also offered (simply titled Live Migration), which allows a virtual machine instance to keep running even during a host system event.

Additional tutorials and a quick start guide are available here.

Read more

- Google Cloud Platform product pricing (Google)
- Google Cloud Platform pricing calculator (Google)
- Google's new Always Free tier gives your company a taste test of public cloud (TechRepublic)
- Microsoft Azure: The smart person's guide (TechRepublic)
- Amazon Web Services: The smart person's guide (TechRepublic)
- Just how big is Google's decision to throw its weight behind OpenStack? (ZDNet)

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